Restoration of images with spatially variant blur by the GMRES method

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ABSTRACT

The GMRES method is a popular iterative method for the solution of linear systems of equations with a large nonsymmetric nonsingular matrix. However, little is known about the performance of the GMRES method when the matrix of the linear system is of ill-determined rank, i.e., when the matrix has many singular values of different orders of magnitude close to the origin. Linear systems with such matrices arise, for instance, in image restoration, when the image to be restored is contaminated by noise and blur. We describe how the GMRES method can be applied to the restoration of such images. The GMRES method is compared to the conjugate gradient method applied to the normal equations associated with the given linear system of equations. The numerical examples show the GMRES method to require less computational work and to give restored images of higher quality than the conjugate gradient method.

Keywords: regularization, ill-posed problem, iterative method, image restoration

1. INTRODUCTION

The restoration of images that have been contaminated by noise and blur can be accomplished by computing an approximate solution of linear system of equations of the form

$$A\tilde{x} = \tilde{b}, \qquad A \in \mathbf{R}^{n\times n}, \qquad \tilde{x}, \tilde{b} \in \mathbf{R}^n, \tag{1}$$

where the available blurred and noisy image is represented by the right-hand side vector $\tilde{b}$; in the computed examples in this paper each entry of $\tilde{b}$ represents a pixel of the image. The matrix A represents the blurring operator. Typically, A has many singular values of different orders of magnitude close to the origin; thus, the matrix A is of ill-determined rank.

Linear systems of equations with matrices of ill-determined rank also arise from the discretization of ill-posed problems, such as Fredholm integral equations of the first kind with a smooth kernel. We refer to linear systems of equations (1) with a matrix of ill-determined rank as discrete ill-posed problems. A recent discussion on many numerical methods for the solution of discrete ill-posed problems is provided by Hansen.[1]

Let $x \in \mathbf{R}^n$ represent the desired, but unkown, "original" blur- and noise-free discretized image, and let the vector $b \in \mathbf{R}^n$ be defined by

$$Ax = b. \tag{2}$$

Thus, b represents the blurred, but noise-free, discretized image associated with x. In many applications of image restoration, the vector b is not available. Instead, the vector

$$\tilde{b} = b + d \tag{3}$$

is known. The entries of the vector $d \in \mathbf{R}^n$ contain measurement and transmission errors. We refer to d as "noise." Thus, the available vector $\tilde{b}$ represents a discretized image, that is contaminated by blur and noise. In the present paper, we assume that the norm of the noise

$$\delta = \|d\| \tag{4}$$

is explicitly known, but that the noise vector d is not. Throughout this paper $\|\cdot\|$ denotes the Euclidean vector norm.

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The desired discretized image x solves the linear systems of equations (2) with unknown right-hand side vector b. We seek to determine an approximation of x by computing an approximate solution of the linear system of equations (1) with known right-hand side (3). Note that since the matrix has many singular values close to the origin, it is severely ill-conditioned. Therefore, the exact solution $\tilde{x}$ of (1), if it exists, typically is a poor approximation of the solution x of (2).

The GMRES method by Saad and Schultz[2] is one of the most popular iterative methods for the solution of linear systems of equations with a large nonsymmetric and nonsingular matrix. However, little is known about the performance of the method when applied to the solution of discrete ill-posed problems. The present paper seeks to shed light on the behavior of the GMRES method when applied to the computation of approximate solutions of large discrete ill-posed problems (1) that arise in image restoration. We consider restoration problems, where the given discrete image $\tilde{b}$ is contaminated by spatially variant blur and noise d of known norm δ; cf. (3) and (4). The number of iterations required by the GMRES method is determined by the discrepancy principle. A detailed discussion of the GMRES method and this termination criterion for the number of iterations is provided in Section 2.

We compare the performance of the GMRES method with that of the Conjugate Gradient method (CG) applied to the iterative solution of the normal equations

$$A^T A\tilde{x} = A^T\tilde{b} \tag{5}$$

associated with the linear system (1). We use the implementation CGLS by Björck[3] of the CG method. This implementation does not require the symmetric positive definite (or semidefinite) matrix $A^T A$ to be formed.

The CG method is an attractive iterative method for the solution of linear systems of equations with a symmetric positive definite matrix; it also can be applied to the solution of consistent linear systems of equations with a symmetric positive semidefinite matrix. Recently, the computation of an approximate solution of (2) by carrying out a few iterations with the CG method applied to the normal equations (5) has received considerable attention. An analysis of this method is given by Hanke[4] and successful applications to the restoration of images degraded by noise and spatially variant blur are reported by Nagy and O'Leary[5,6]; see also Björck[3] and Hanke and Hansen[7] for discussions of the CGLS method.

Since the singular values of the matrix $A^T A$ are the square of the singular values of A, it follows that if A has many singular values of different orders of magnitude close to the origin, then so does $A^T A$. Hence, the matrix $A^T A$ is of ill-determined rank when A is, and the linear system of equations (5) is a discrete ill-posed problem whenever the linear system (1) is.

Each iteration by the CGLS method applied to (5) requires two matrix-vector product evaluations, one with the matrix A and one with A^T. For comparison, we note that each iteration by the GMRES method applied to (1) requires only one matrix-vector product evaluation.

It may be possible to reduce the number of iterations required by using a preconditioner. However, the determination of a suitable preconditioner for discrete ill-posed problems of the form (1) or (5) is generally not straightforward. Therefore, in order to keep the comparison simple, we do not precondition the linear systems of equations in the computed examples reported in Section 3. The examples show the GMRES method applied to (1) to require less computational work and give restored images of higher quality than the CGLS method applied to the normal equations (5).

An analysis of the GMRES method with the stopping criterion of the present paper is presented by Calvetti et al..[8] Other stopping criteria, that do not require the norm of the noise δ to be known, are discussed by Calvetti et al..[9]

2. THE GMRES METHOD AND THE DISCREPANCY PRINCIPLE

We discuss the computation of an approximate solution of the linear system of equations (2) with the unknown right-hand side vector b, by applying the GMRES method to the linear system of equations (1) with the known right-hand side vector (3). Throughout this paper, we apply the GMRES method with initial approximate solution $\tilde{x}_0 = 0$.

We first review properties of the GMRES method. For further details and extensions of the method, we refer to the work by Saad and Schultz.[2,10] Introduce the Krylov subspaces

$$\mathcal{K}_m(A,\tilde{b}) = \text{span}\{\tilde{b}, A\tilde{b}, \ldots, A^{m-1}\tilde{b}\}, \qquad m = 1,2,3,\ldots\ ,$$

and note that

$$\mathcal{K}_m(A,\tilde{b}) \subset \mathcal{K}_{m+1}(A,\tilde{b}), \qquad m = 1,2,3,\ldots\ . \tag{6}$$

Denote the approximate solution determined by applying m steps of the GMRES method to the linear system of equations (1) by $\tilde{x}_m$. Then $\tilde{x}_m$ satisfies

$$\|A\tilde{x}_m - \tilde{b}\| = \min_{\tilde{x}\in\mathcal{K}_m(A,\tilde{b})} \|A\tilde{x} - \tilde{b}\|, \qquad \tilde{x}_m \in \mathcal{K}_m(A,\tilde{b}). \tag{7}$$

It follows from (6) and (7) that the approximate solutions determined by the GMRES method satisfy

$$\|A\tilde{x}_m - \tilde{b}\| \leq \|A\tilde{x}_{m-1} - \tilde{b}\|, \qquad m = 1,2,3,\ldots\ . \tag{8}$$

Let $\tilde{b} \neq 0$ and assume for the moment that the matrix A is nonsingular. Substituting the Jordan canonical form of A into (7) shows that there is a unique integer ℓ, such that

$$A\tilde{x}_\ell = \tilde{b}, \qquad A\tilde{x}_{\ell-1} \neq \tilde{b}, \qquad 1 \leq \ell \leq n. \tag{9}$$

In applications of the GMRES method to the approximate solution of discrete ill-posed problems, typically many fewer than ℓ steps of the method are carried out.

In the present paper, we terminate the iterations as soon as an approximate solution $\tilde{x}_m$ has been determined, such that the associated residual error $A\tilde{x}_m - \tilde{b}$ is of norm smaller than or equal to δ, where δ is the norm of the noise in the right-hand side vector $\tilde{b}$; cf. (3) and (4). Thus, we terminate the computations as soon as an approximate solution $\tilde{x}_m$ has been determined, such that

$$\|A\tilde{x}_m - \tilde{b}\| \leq \delta, \qquad \|A\tilde{x}_{m-1} - \tilde{b}\| > \delta. \tag{10}$$

The residual error $A\tilde{x}_m - \tilde{b}$ is sometimes referred to as the discrepancy and this stopping criterion as the discrepancy principle; see Hansen[1] or Morozov[12] for a discussion of its properties.

The Arnoldi decompositions

$$AV_m = V_{m+1}\bar{H}_m, \qquad m = 1,2,3,\ldots\ , \tag{11}$$

form the basis of the implementation of the GMRES method proposed by Saad and Schultz.[2] The matrices V_m in (11) satisfy

$$V_m \in \mathbf{R}^{n\times m}, \qquad V_m^T V_m = I_m, \qquad \text{range}(V_m) = \mathcal{K}_m(A,\tilde{b}), \qquad V_m e_1 = \tilde{b}/\|\tilde{b}\|, \tag{12}$$

where I_m denotes the $m \times m$ identity matrix and e_1 the first axis vector. The matrices $\bar{H}_m = [h_{ij}] \in \mathbf{R}^{(m+1)\times m}$ are of upper Hessenberg form; all entries h_{ij} with $i > j+1$ vanish. Moreover, the subdiagonal entries $h_{j+1,j}$ are positive. It can be shown that if A is nonsingular, then Arnoldi decompositions (11) with the stated properties exist for all $m < \ell$, where ℓ is determined by (9).

Let $\beta = \|\tilde{b}\|$, assume that A is nonsingular and let the index ℓ be determined by (9). Then equations (7), (11) and (12) yield, for $m \leq \ell$,

$$\|A\tilde{x}_m - \tilde{b}\| = \min_{\tilde{x}\in\mathcal{K}_m(A,\tilde{b})} \|A\tilde{x} - \tilde{b}\| = \min_{\tilde{y}\in\mathbf{R}^m} \|AV_m\tilde{y} - \tilde{b}\| \tag{13}$$

$$= \min_{\tilde{y}\in\mathbf{R}^m} \|V_{m+1}(\bar{H}_m\tilde{y} - \beta e_1)\| = \min_{\tilde{y}\in\mathbf{R}^m} \|\bar{H}_m\tilde{y} - \beta e_1\|. \tag{14}$$

Thus, the minimization problem (7) with the large matrix A has been replaced by the minimization problem in the right-hand side of (14) with the generally much smaller matrix $\bar{H}_m$. Denote the solution of the latter minimization problem by $\tilde{y}_m$. Then

$$\tilde{x}_m = V_m\tilde{y}_m. \tag{15}$$

Introduce the QR-factorization

$$\bar{H}_m = Q_{m+1}\bar{R}_m, \tag{16}$$

where $Q_{m+1} \in \mathbf{R}^{(m+1)\times(m+1)}$ satisfies $Q_{m+1}^T Q_{m+1} = I_{m+1}$ and the leading $m \times m$ submatrix of $\bar{R}_m \in \mathbf{R}^{(m+1)\times m}$ is upper triangular; the last row of $\bar{R}_m$ has only zero entries. Then

$$\min_{\tilde{y}\in\mathbf{R}^m} \|\bar{H}_m\tilde{y} - \beta e_1\| = \min_{\tilde{y}\in\mathbf{R}^m} \|\bar{R}_m\tilde{y} - \beta Q_{m+1}^T e_1\| = \beta e_{m+1}^T Q_{m+1}^T e_1. \tag{17}$$

Introduce the quantities

$$\rho_m = \beta e_{m+1}^T Q_{m+1}^T e_1, \qquad m = 1, 2, 3, \ldots .$$

It follows from (13)-(14) that

$$\rho_m = \|A\tilde{x}_m - \tilde{b}\|. \tag{18}$$

We are in a position to describe the GMRES implementation by Saad and Schultz[2] with stopping criterion (10).

ALGORITHM 2.1. *The GMRES method with stopping criterion (10)*

Input: A, $\tilde{b}$, δ.

Output: Approximate solution $\tilde{x}_m$ *of (2) that satisfies (7) and (10).*

$\beta := \|\tilde{b}\|$, $v_1 := \tilde{b}/\beta$, $m := 0$, $\rho_0 := \beta$,

while $\rho_m > \delta$ *do*

  $m := m + 1$, $w := Av_m$,
  for $i = 1, 2, \ldots, m$ *do*
    $h_{im} := v_i^T w$, $w := w - h_{im} v_i$,
  end
  $h_{m+1,m} := \|w\|$, $v_{m+1} := w/h_{m+1,m}$,
  for $i = 1, 2, \ldots, m-1$ *do*
    $h_{m+1,i} := 0$,
  end
  $\bar{H}_m := [h_{i\ell}]_{1\le i\le m+1, 1\le \ell\le m}$,
  % $\bar{H}_m$ is the Hessenberg matrix in the Arnoldi decomposition (11).
  Compute the QR-factorization (16) of $\bar{H}_m$,
  $\rho_m := \beta e_{m+1}^T Q_{m+1}^T e_1$,

end

if $m = 0$ *then*

  $\tilde{x}_0 := 0$,

else

  $V_m := [v_1, v_2, \ldots, v_m]$,
  Solve the minimization problem (13) using (17). Denote the solution by $\tilde{y}_m$ *and compute* $\tilde{x}_m$ *according to (15).*

end

The matrix Q_{m+1} in the QR-factorization of $\bar{H}_m$ is conveniently represented in terms of m Givens rotations. Since $\bar{H}_{m-1}$ is a submatrix of $\bar{H}_m$, the QR-factorization of the latter matrix can be computed inexpensively by updating the QR-factorization of the former.

The quantity ρ_m computed in the algorithm satisfies (18) in the absence of round-off errors. In the computed examples of the present paper, only fairly few iterations with the GMRES method are required and, therefore, the effect of round-off errors can be ignored.

3. APPLICATION TO IMAGE RESTORATION

We consider image restoration problems obtained from the Space Telescope Science Institute. The files that represent the discrete blurred and noisy image and the point-spread functions that are used to construct the blurring operators, as well as a file that represents the "original" discrete noise- and blur-free image, can be obtained by ftp from the address *ftp.stsci.edu* in the directory */pub/software/stsdas/testdata/restore/sims/star_cluster*. The computational task is to restore noisy blurred astronomical images that simulate the spatially variant blur associated with the Hubble Space Telescope's original (defective) imaging system. The images are represented by 256×256 arrays of floating-point numbers in the interval $[2.6 \cdot 10^{-11}, 3.2 \cdot 10^4]$. These numbers represent grayscale values of pixels of the image. An "original" noise- and blur-free image is available at the ftp site in the file *truth.fit*, stored in the fits format. This image contains 470 point-sources ("stars") and is referred to as the model image. A discrete blurred and noisy version of this image is available at the ftp site in the file *sim3.fit*. This image was generated by first convolving the model image with point-spread functions, and then adding read-out noise and Poisson noise. Details on the construction of the blurred and noisy image are described in the README file at the ftp site. Following Nagy and O'Leary,[5] we truncate each grayscale value p of each pixel of the discrete blurred and noisy image to $\min(\max(p, 50), 500)$ in order to avoid oversaturation in the displayed pictures. In our display of astronomical images, the pixel value zero corresponds to black and the value 500 to white.

The truncated grayscale values of the pixels of the blurred and noisy image are stored row-wise in the right-hand side vector $\tilde{b}$ with $n = 256^2$ entries. Analogously, the grayscale values of the pixels of the "original" noise- and blur-free model image are stored row-wise in the vector x. The model image as well as the blurred and noisy image are displayed in Figure 2.

We model spatially variant blur as follows. Partition the discrete image into s non-overlapping subimages and let the blur in each subimage be spatially invariant. The data set at the above mentioned ftp site includes 25 discrete point-spread functions, each one inducing a spatially invariant, non-separable blurring operator. The matrix A in (1) represents the blurring operator for the entire image. It can be written in the form

$$A = D_1 A_1 + D_2 A_2 + \cdots + D_s A_s, \tag{19}$$

where the matrix A_j represents the blurring operator induced by the point-spread function for the jth subimage and each D_j is a diagonal matrix, whose ith diagonal entry is 1 when the ith pixel of the full image is in the jth subimage; otherwise the ith diagonal entry is set to zero.

Thus, given the blurred and noisy image shown in Figure 2, we seek to determine a good approximation of the model image, also shown in Figure 2, by solving systems of equations (1) with blurring operators A determined by (19) for several values of s. The number of point-spread functions s is chosen to balance accuracy in the representation of the true blurring operator and computational efficiency.

We present examples computed with $1, 4$ and 25 point-spread functions. The partitioning of the image into 4 and 25 subimages, respectively, is displayed by Figure 1. When $s = 4$, the image is divided into four subimages, each of which is represented by an array of 128×128 pixels values. Subdivision of the image into 25 subimages is slightly more complicated. All subimages, except for those along the bottom or right-hand side edges are represented by arrays of 51×51 pixels. The subimages along the bottom edge have an extra row of pixels and the subimages along the right-hand side edge contain an extra column of pixels. Each one of the 25 point-spread functions is stored in a 60×60 array. The point-spread functions are labeled MPSF00 to MPSF24. Figure 1 displays how the point-spread functions are associated with the subimages. For instance, the size of the subimage associated with the point-spread function MPSF10 is 52×51 pixels. A linear transformation is applied to each point-spread function array, so that the minimum entry in the array is zero and the entries sum to one. The matrices A in the computed examples below, determined by 1, 4 and 25 point-spread functions, respectively, are nonsymmetric.

To estimate the norm of the noise in $\tilde{b}$, we compute $\delta = \|Ax - \tilde{b}\|$, where A is the blurring operator constructed with 25 point-spread functions in (19). The value $\delta = 512.6$ so obtained corresponds to a noise level of 7.1% relative to $\|Ax\|$. We remark that $\|x\|/\|Ax\| = 7.1$.

All computations were carried out using Octave 2.0.16 on a cluster of Intel Pentium workstations with floating-point arithmetic with 16 significant digits. We used the program fv to convert the arrays stored in the fits format at the ftp site to floating-point arrays for input to Octave. The floating-point representations of the images was converted to PostScript by the Gnu image manipulation program.

Formula (19) splits the evaluation of a matrix-vector product with the matrix A naturally into the evaluation of s independent matrix-vector products with the matrices $D_j A_j$. This property is well suited for parallel implementations of the GMRES and CGLS methods on a cluster of workstations.

The computed examples compare the GMRES method applied to the linear system (1) to the conjugate gradient method CGLS applied to the normal equations (5). We use the initial approximate solution $\tilde{x}_0 = 0$ in all examples both for the CGLS and GMRES methods. Since the model image x is available, we can evaluate the relative error norm $\|\tilde{x}_m - x\|/\|x\|$ for the iterates determined by the GMRES and CGLS methods. We also evaluate the residual error norm $\|A\tilde{x}_m - \tilde{b}\|$ associated with the computed iterates.

Example 1. The blurring operator A is defined by one spatially invariant point-spread function, MPSF11. We perform 40 steps of the GMRES method and 50 steps of the CGLS method. Figure 7 displays the relative error norms and residual error norms for the approximate solutions $\tilde{x}_m$ determined by the GMRES and CGLS methods. The points corresponding to each iteration in this and analogous graphs are connected by straight lines for clarity. The minimum relative error norm achieved by the GMRES method is $1.8 \cdot 10^{-2}$ at step 9, and the minimum relative error norm achieved by the CGLS method is $1.8 \cdot 10^{-2}$ at step 44.

Figure 3 shows the restored images determined by the GMRES and CGLS methods when the iterations were terminated according to the discrepancy principle (10). For the GMRES method this stopping criterion is satisfied after 7 steps, for the CGLS method after 24 steps. We remark that the computation of 7 steps of the GMRES method requires the evaluation of 7 matrix-vector products with the matrix A, while the the computation of 24 steps of the CGLS method requires the evaluation of 48 matrix-vector products, 24 with A and 24 with A^T. The restored image determined by the GMRES method displays less ghosting than the restored image determined by the CGLS method. □

Example 2. The blurring operator A is defined by four spatially invariant point-spread functions. The subdivision of the image and the names of the point-spread functions used in each subimage are displayed in Figure 1. We carried out 25 steps of the GMRES method and 50 steps of the CGLS method. Figure 8 compares the relative error norms and the residual error norms for the approximate solutions $\tilde{x}_m$ determined by the GMRES and CGLS methods. The minimum relative error norm achieved by both methods is $1.6 \cdot 10^{-2}$. This value is attained at step 11 of the GMRES method and at step 48 of the CGLS method.

Figure 4 displays the restored images determined by the GMRES and CGLS methods when the iterations are terminated according to the discrepancy principle (10). For the GMRES method this stopping criterion is satisfied after 7 steps, for the CGLS method after 25 steps. Similarly as in Example 1, the restored image determined by the GMRES method displays less ghosting than the restored image computed by the CGLS method. □

Example 3. The blurring operator A is defined by 25 spatially invariant point-spread functions. The subdivision of the image and the names of the point-spread functions used are displayed in Figure 1. We carried out 20 steps of GMRES method and 50 steps of the CGLS method. Figure 9 compares the relative error norms and the residual error norms for the approximate solutions determined by the GMRES and CGLS methods. The minimum relative error norm achieved by the GMRES method is $1.6 \cdot 10^{-2}$ at step 11, and the CGLS method achieves the minimum relative error norm $1.6 \cdot 10^{-2}$ at step 50.

Figure 5 displays the restored images determined by the GMRES and CGLS methods when the iterations are terminated according to the discrepancy principle (10). For the GMRES method this stopping criterion is satisfied after 7 steps, for the CGLS method after 25 steps. Figure 6 displays the modulus of the difference of the model image of Figure 2 and the restored images of Figure 5. □

Table 1 summarizes key numerical quantities for the examples. The table and the figures show the restored images determined by the GMRES method with stopping criterion (10) to better approximate the model image than the restored images determined by the CGLS method with the same stopping criterion.

4. ACKNOWLEDGEMENTS

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	Example 1		Example 2		Example 3	
	GMRES	CGLS	GMRES	CGLS	GMRES	CGLS
m	7	24	7	25	7	25
$\|\tilde{b} - A\tilde{x}_m\|$	$3.5 \cdot 10^2$	$5.1 \cdot 10^2$	$3.6 \cdot 10^2$	$5.0 \cdot 10^2$	$3.6 \cdot 10^2$	$5.0 \cdot 10^2$
$\|\tilde{x}_m - x\|/\|x\|$	$1.9 \cdot 10^{-1}$	$2.1 \cdot 10^{-1}$	$1.7 \cdot 10^{-1}$	$2.0 \cdot 10^{-1}$	$1.7 \cdot 10^{-1}$	$2.0 \cdot 10^{-1}$

Table 1. Numerical results for Examples 1-3.

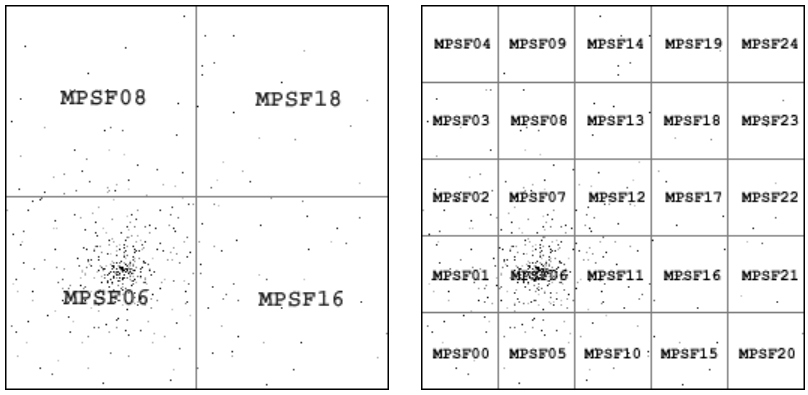

Figure 1. Image partitions and associated point-spread functions.

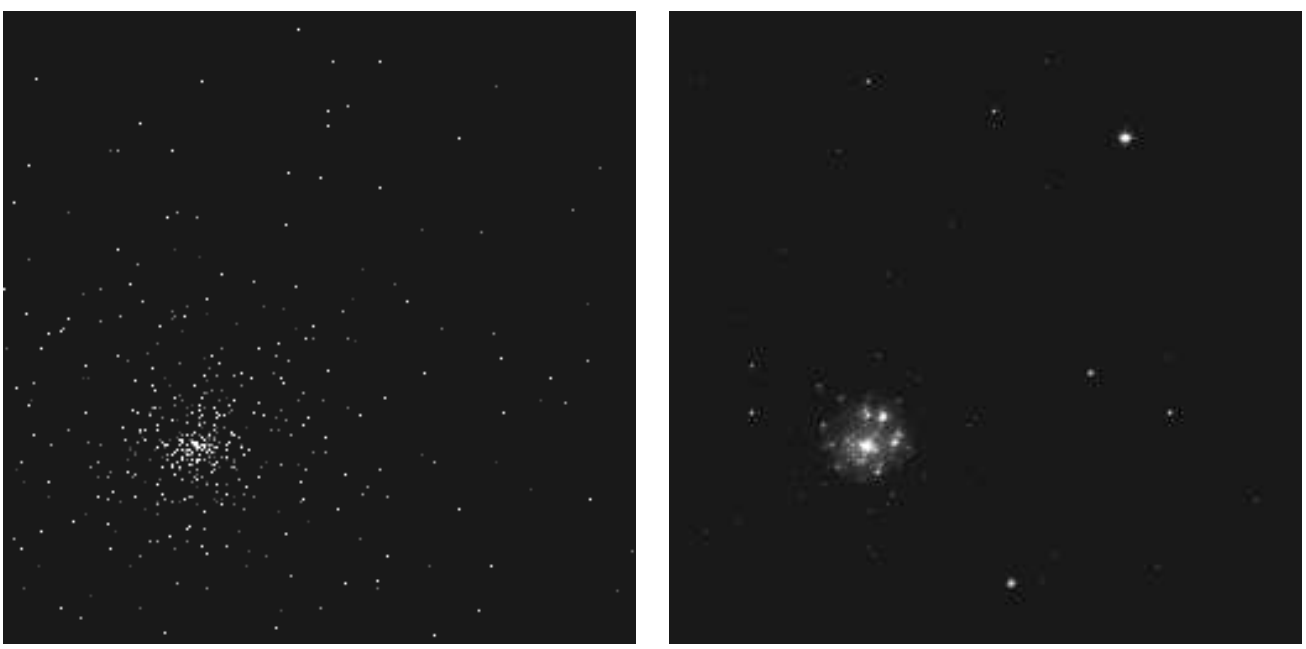

Figure 2. Blur- and noise-free model image (left), blurred and noisy image (right).

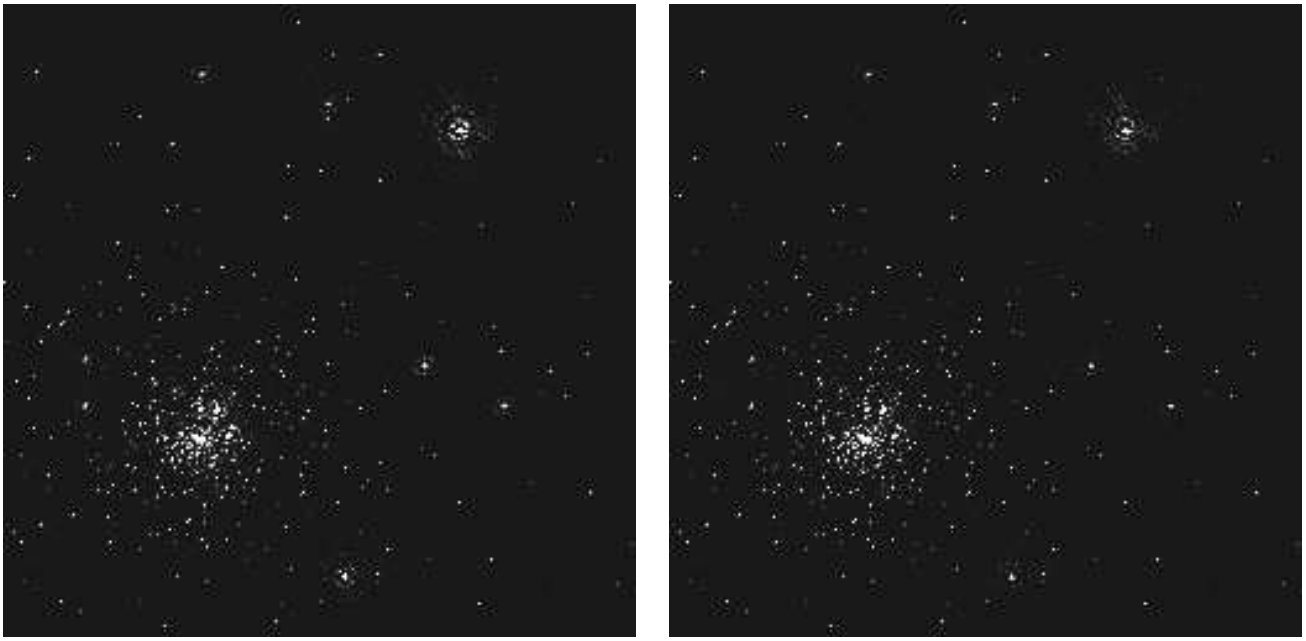

Figure 3. Example 1: Restored images using one point-spread function and the stopping criterion (10). The iterations were terminated after 24 steps of the CGLS method (left-hand side image) and after 7 steps of the GMRES method (right-hand side image).

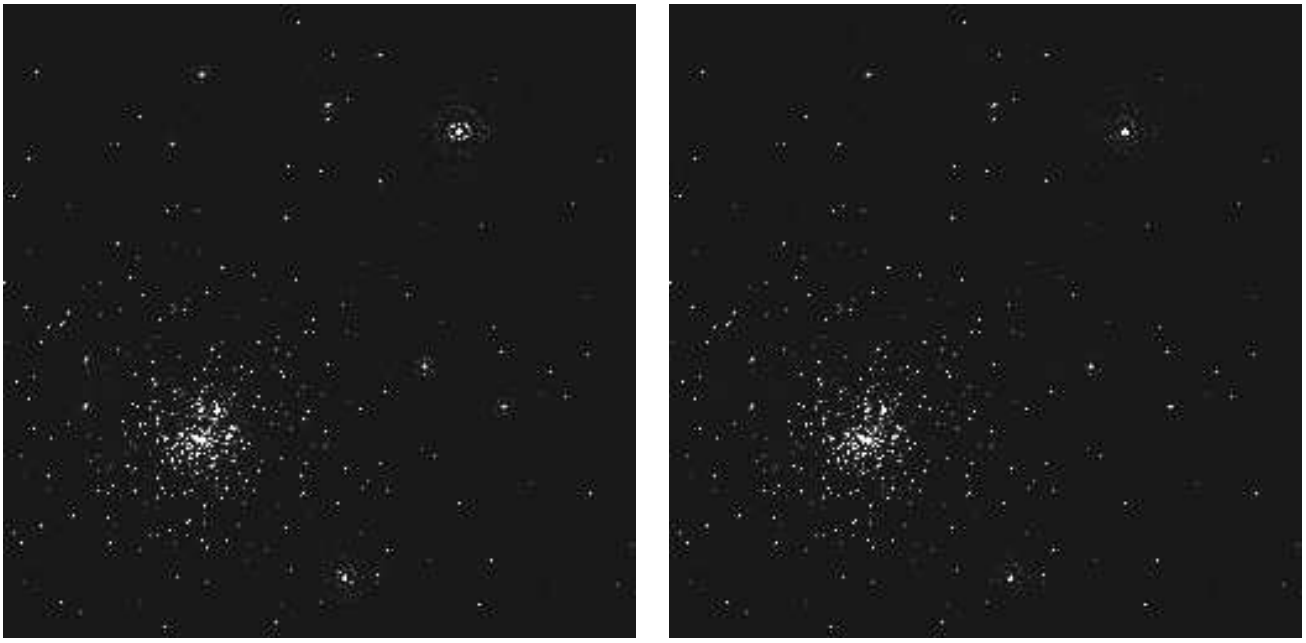

Figure 4. Example 2: Restored images using four point-spread function and the stopping criterion (10). The iterations were terminated after 25 steps of the CGLS method (left-hand side image) and after 7 steps of the GMRES method (right-hand side image).

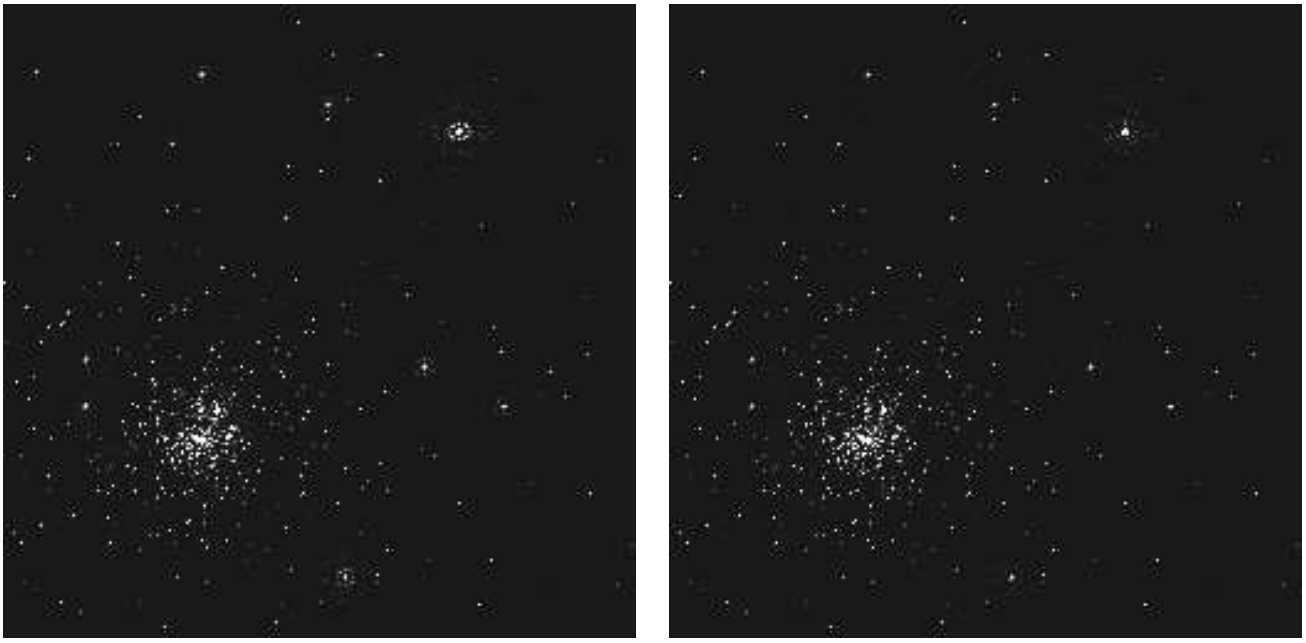

Figure 5. Example 3: Restored images using 25 point-spread function and the stopping criterion (10). The iterations were terminated after 25 steps of the CGLS method (left-hand side image) and after 7 steps of the GMRES method (right-hand side image).

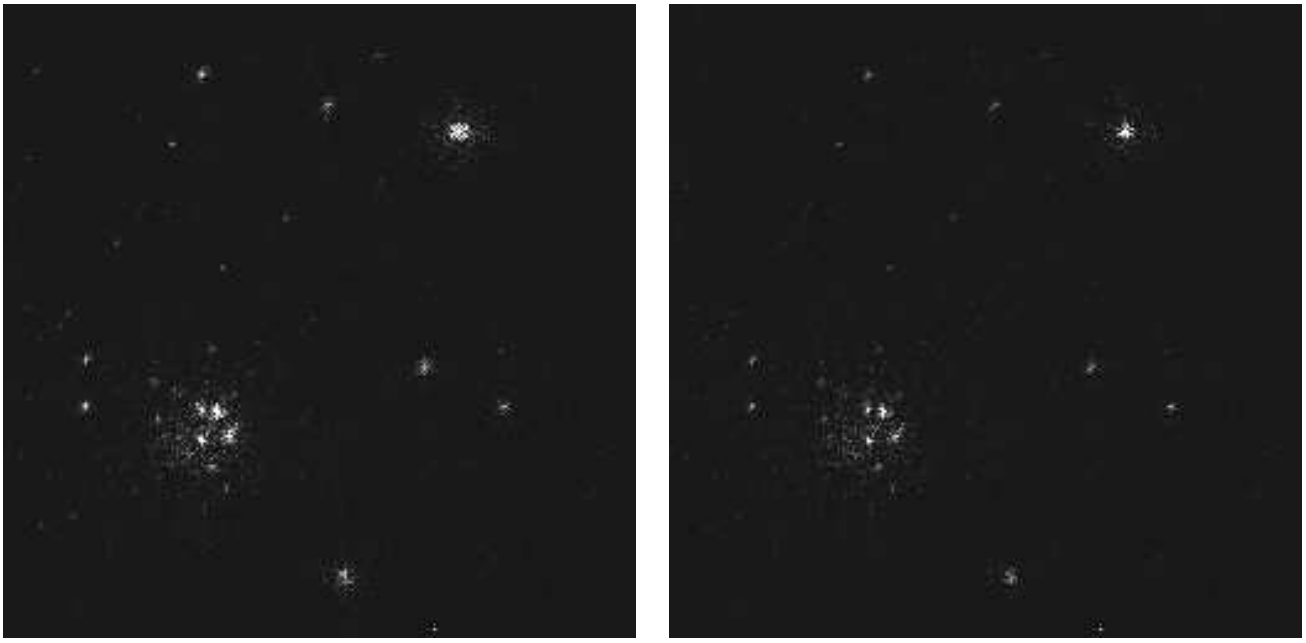

Figure 6. Example 3: Difference images for the approximate solutions determined after 25 steps of the CGLS method (left-hand side image), and after 7 steps of the GMRES method (right-hand side image).

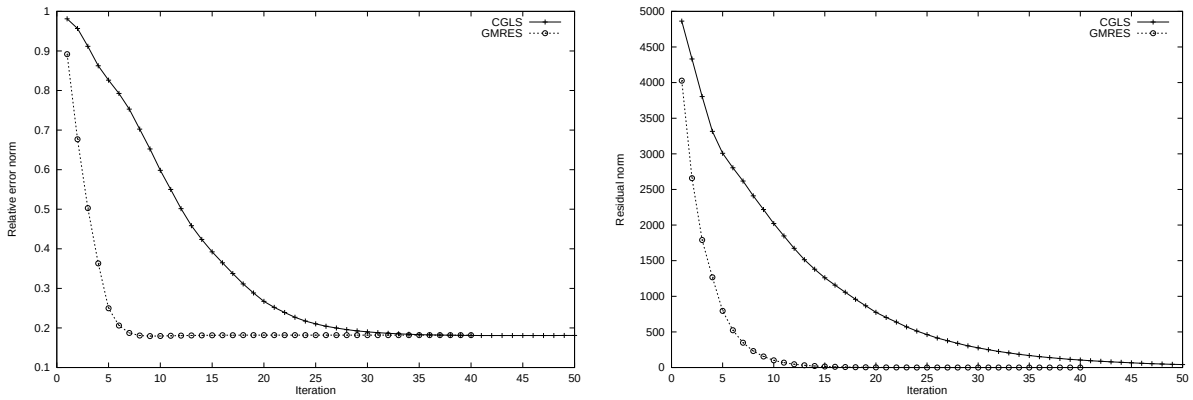

Figure 7. Example 1: Comparison of the CGLS and GMRES methods

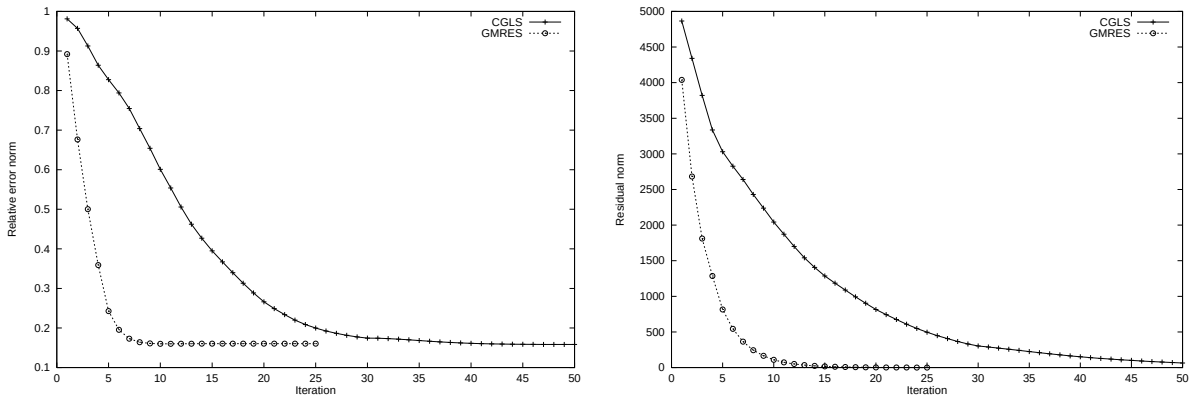

Figure 8. Example 2: Comparison of the CGLS and GMRES methods

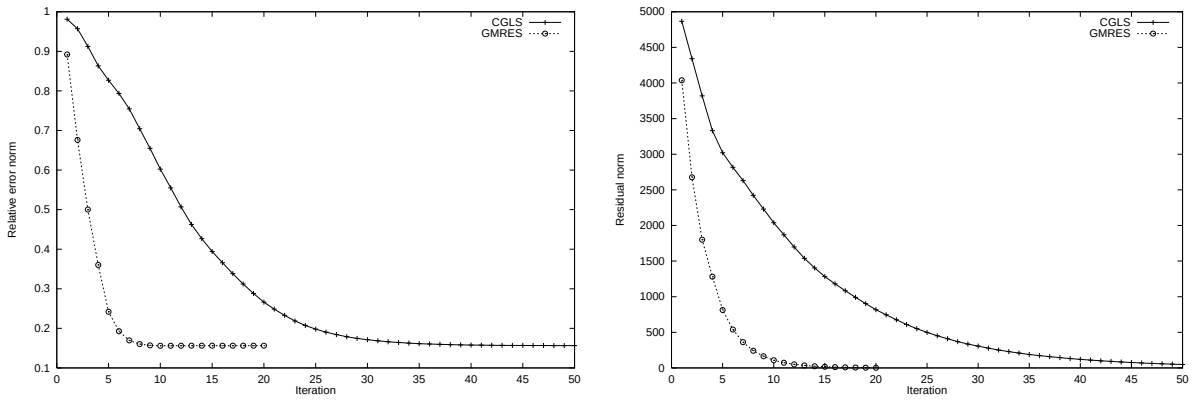

Figure 9. Example 3: Comparison of the CGLS and GMRES methods

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